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nguseyxc92

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Neguse Solomon Mekonnen

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Engineering • Technology • Management

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Tebib

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nguseyxc92/README.md

Neguse Solomon Mekonnen 👋

Engineering • Technology • Research

I am a Civil & Software Engineer and Researcher dedicated to the intersection of Structural Engineering, Computer Science, and Data-Driven Systems. I combine engineering principles, Bayesian reliability, and modern software architecture to design, analyze, and build intelligent, resilient systems—ranging from physical bridge infrastructures to scalable digital tools.

🛠️ Professional Focus

Structural Engineering & Reliability: Utilizing computational mechanics, risk assessment, and Bayesian inference to optimize infrastructure and manage structural lifecycles.

Software Architecture & Scientific Programming: Crafting full-stack applications, data pipelines, and AI-driven solutions that transform complex data into actionable engineering tools.

Research & Innovation: Developing deep-learning surrogate twins (e.g., for urban drainage systems), bridge reliability frameworks, and proof load testing methodologies.

🔬 Flagship Research & Development

1. Bayesian Reliability Updating of Bridge Structural Reliability Using Monitoring Data and Stop Criteria: A Case Study of the Zijlweg Viaduct

A state-of-the-art framework for quantifying bridge reliability using field monitoring data.

Tech Stack: C++, Python (Data Science), MATLAB.

Impact: Integrating AI-driven probabilistic modeling with structural analysis to ensure safety and optimize maintenance.

2. Deep-Learning Surrogate Digital Twins for Predictive Capacity Auditing and Maintenance Optimization

An advanced ML framework creating high-speed surrogate models for large-scale hydraulic infrastructure reliability tracking.

Case Study: Jemo–Lancha box-culvert corridor, Nifas Silk-Lafto, Addis Ababa.

Tech Stack: Python (Scikit-Learn, PySWMM, NumPy, SciPy), EPA SWMM 5.

Impact: Integrates Latin Hypercube Sampling (LHS) and ANN/GPR ensembles to run safety analysis 8,000× faster than dynamic-wave models, providing optimized condition-based maintenance thresholds.

🚀 Featured Projects

1. Tebib Labs: Official Website

The corporate website of Tebib Labs, a software engineering and applied R&D company. It includes interactive Bayesian reliability and sensor-telemetry visualizations, a technical insights journal and project pages.

Live: tebib-labs.netlify.app · GitHub Pages

Stack: React · TypeScript · Vite · Tailwind CSS

💻 Tech Ecosystem

Infrastructure & Engineering Software & AI

Analysis: ETABS, SAP2000, SAFE Languages: Python, JavaScript, C++

Design: AutoCAD, Civil 3D, Revit Web: React, Node.js, Next.js

BIM: Structural Design & Planning AI/ML: NumPy, SciPy, MCMC

📊 GitHub Activity

Neguse's Stats GitHub Streak

Note: If the stats card above is not visible, it is due to global server traffic. For a 100% permanent solution, I recommend using the GitHub Metrics action below.

📬 Let's Build the Future

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Always Dream!

"Nobody's ever too old to set another goal, or to dream another dream."

— C.S. Lewis

PinnedLoading

zijlweg-bayesian-reliability Public

Bayesian Reliability Updating Framework for Bridge Structures using C++, Python, and MATLAB. Case study on the Zijlweg Viaduct (Netherlands).

C++ 2 1

51 contributions in the last year

Skip to contributions year list

Contribution Graph

Day of Week NovemberNov DecemberDec JanuaryJan FebruaryFeb MarchMar AprilApr MayMay JuneJun JulyJul AugustAug SeptemberSep OctoberOct NovemberNov

SundaySun

MondayMon

TuesdayTue

WednesdayWed

ThursdayThu

FridayFri

SaturdaySat

Learn how we count contributions

Less

No contributions.

Low contributions.

Medium-low contributions.

Medium-high contributions.

High contributions.

More

Contribution activity

November 2026

nguseyxc92/nguseyxc92 11 commits

nguseyxc92/tebib_labs 6 commits

nguseyxc92/gpc-ethiopia-assessment 1 commit

nguseyxc92/tebib_labs

TypeScript

This contribution was made on Nov 9Nov 9

nguseyxc92/gpc-ethiopia-assessment

Python

This contribution was made on Nov 5Nov 5

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2026

2025

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Special repository containing my personal profile, academic engineering research, software development portfolio, and background as a Civil & Software Engineer.

Updated 2 hours ago

tebib_labs Public

Official website of Tebib Labs, a software engineering and applied R&D company building computational tools for structural reliability, Bayesian analysis and scientific computing. Built with React,…

TypeScript Updated 3 hours ago

gpc-ethiopia-assessment Public

A literature-calibrated assessment of carbon, cost, and structural reliability comparing natural-pozzolan geopolymer concrete against local PPC and OPC mixes in Ethiopia using Monte Carlo simulatio…

Python MIT License Updated 4 days ago

culvert_twin Public

Deep-learning surrogate digital twins for EPA SWMM 5 predictive capacity auditing and condition-based maintenance optimization. Applied to the Jemo–Lancha box-culvert corridor in Addis Ababa using …

Python MIT License Updated last week

zijlweg-bayesian-reliability Public

Bayesian Reliability Updating Framework for Bridge Structures using C++, Python, and MATLAB. Case study on the Zijlweg Viaduct (Netherlands).

C++ 2 1 Updated last month

https://github.com/nguseyxc92
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